EIP-130 Kernel-space Driver Software Block Diagram

Application.ko

Customer Software

Linux OS

Driver API

Driver API
(api_driver130_init.h)

VAL API
(api_val.h)

LKM EIP-130 Adapter Module
(adapter_*.*)

Driver Entry Point

VAL

Adapter Initialization

VEX

AIC-201 Control

EIP-130 Control

Buffer Manager

Configuration

AIC-201 Driver Library
(eip201_*.*)

EIP-130 Driver Library
(eip130_*.*)

Adapter Configuration
(cs_adapter.h)

Driver Framework DMA resource API
(dmares_*.h)

Driver Framework Device API
(device_*.h)

Driver Framework DMA resource implemetation
(dmares_lkm.c)

Logging for Linux kernel-space
(log*.*)

Driver Framework Device API Implementation for Linux Kernel
(device_lkm.c)

used by all modules

used by many modules

Driver Framework Basic Definitions API
(basic_defs.h)

Driver Framework Clib Abstraction API
(clib.h)

Driver Configuration
(cs_*.h)

Linux Kernel API

C Compiler & C Run-Time Library

Legend

DDK Kit/Integration Component

DDK Example Component

Implementation Module

Sub-module

LKM - Linux loadable Kernel Module

"API" = Interface only

Dependency

Implementation

Rambus

EIP-130 Driver v3.1

item: Kernel Driver Block Diagram

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EIP-130 User-space Driver Software Block Diagram

Linux user process (application)

Linux OS

Customer Software

Driver API

Driver API
(api_driver130_init.h)

VAL API
(api_val.h)

driver_vip_u.a

UM EIP-130 Adapter Module
(adapter_*.*)

Driver Entry Point

VAL

Adapter Initialization

VEX

AIC-201 Control

EIP-130 Control

Buffer Manager

Configuration

AIC-201 Driver Library
(eip201_*.*)

EIP-130 Driver Library
(eip130_*.*)

Adapter Configuration
(cs_adapter.h)

Logging for Linux user-space
(log*.*)

Driver Framework Device API
(device_*.h)

Driver Framework DMA resource API
(dmares_*.h)

Driver Framework Device API Implementation for Linux User Mode
(hwpal_device_umdevxs.c)

Driver Framework DMA resource implemetation
(hwpal_dmares*.c)

UMDevXS Proxy
(umdevxsproxy_*.*)

used by all modules

used by many modules

Driver Framework Basic Definitions API
(basic_defs.h)

Driver Framework Clib Abstraction API
(clib.h)

Driver Configuration
(cs_*.h)

Linux User API

Legend

DDK Kit Component

DDK Integration Component

Implementation Module

Sub-module

LKM - Linux loadable Kernel Module

"API" = Interface only

Dependency

Implementation

Rambus

EIP-130 Driver v3.1

User-space Driver Block Diagram

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EIP-130 Adapter Kernel-space Block Diagram

Application.ko

Application_up

VAL API
(api_val.h)

Adapter VAL
(adapter_val_*.c)

Adapter
VEX API
(adapter_vex_*.h)

VEX Proxy
(vex_proxy.c)

EIP-130 Adapter Kernel

Driver VAL Init API
(api_driver_val_init.h)

Entry Point
(adapter_driver_val_init.c)

LKM Support
(adapter_driver_val_init_ext.h)

Adapter Init API
(adapter_init.h)

Init
(adapter_init.c)

VEX Stub
(vex_stub.c)

Adapter VEX
(adapter_vex.c)

Firmware
API
(adapter_firmware.h)

Firmware
(adapter_firmware.c)

Buffer Manager
API
(adapter_bufmanager.h)

Buffer Manager
(adapter_bufmanager.c)

used by all sub modules

Default Configuration
(c_adapder.h)

Linux Kernel API

Driver Framework
DMAResource API
(dmares_*.h)

AIC-201 Driver Library
API
(eip201_*.h)

EIP-130 Driver Library
API
(eip130_*.h & eip130token_*.h)

External Adapter
Configuration
(cs_adapter.h)

Driver Configuration
(cs_adapter.h)

used by many modules

Logging API
(log.h)

used by many modules

Clib Abstraction API
(clib.h)

used by all modules

Basic Definitions API
(basic_defs.h)

Legend

API — Driver External Interface

API — Driver Internal Interface

Implementation Module — Sub-module

Application

"API" = Interface only

Dependency

Implementation

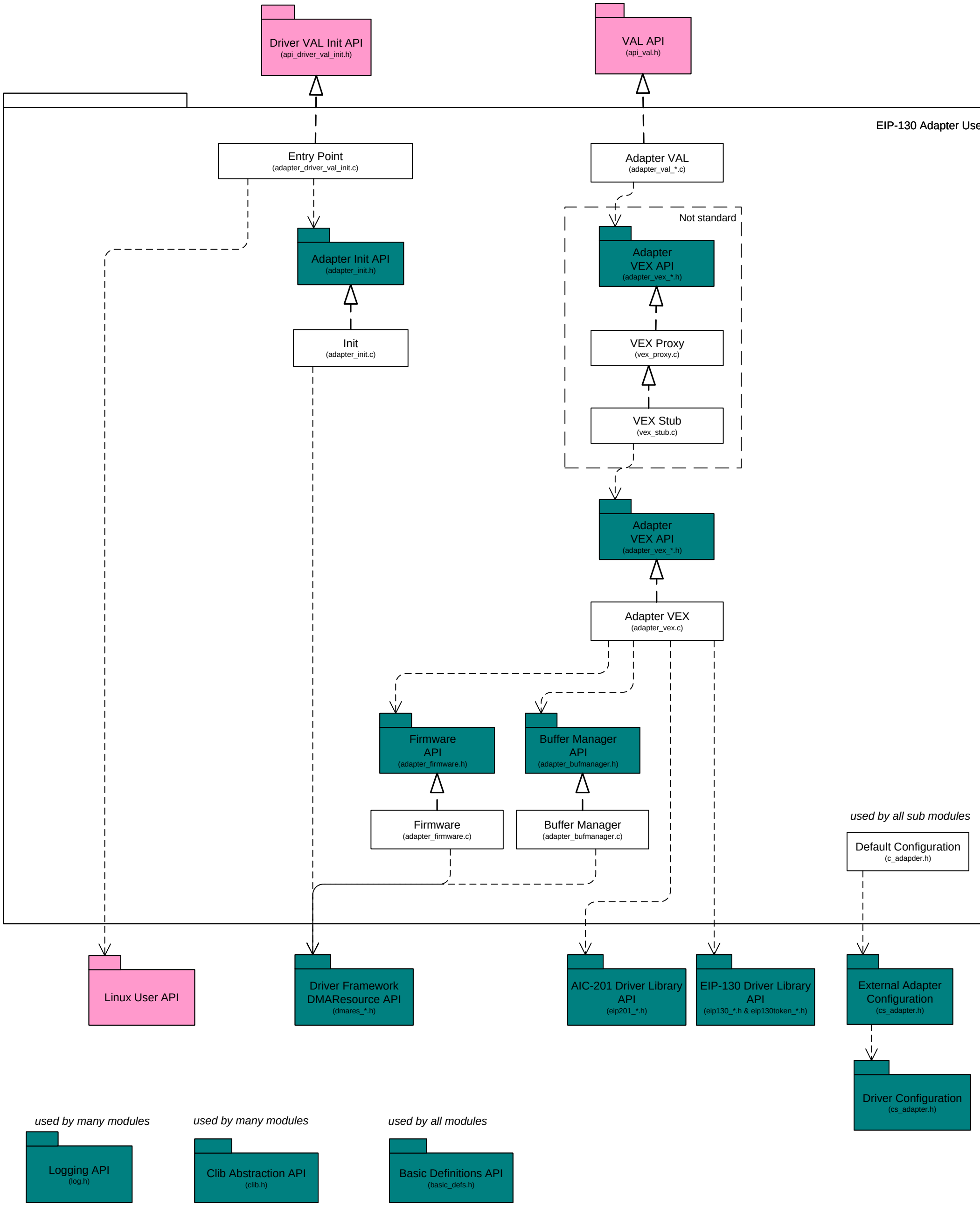
Rambus

EIP-130 Driver v3.1

item: EIP-130 Adapter Kernel Block Diagram

date: 2022-02-03

EIP-130 Adapter User-space Block Diagram

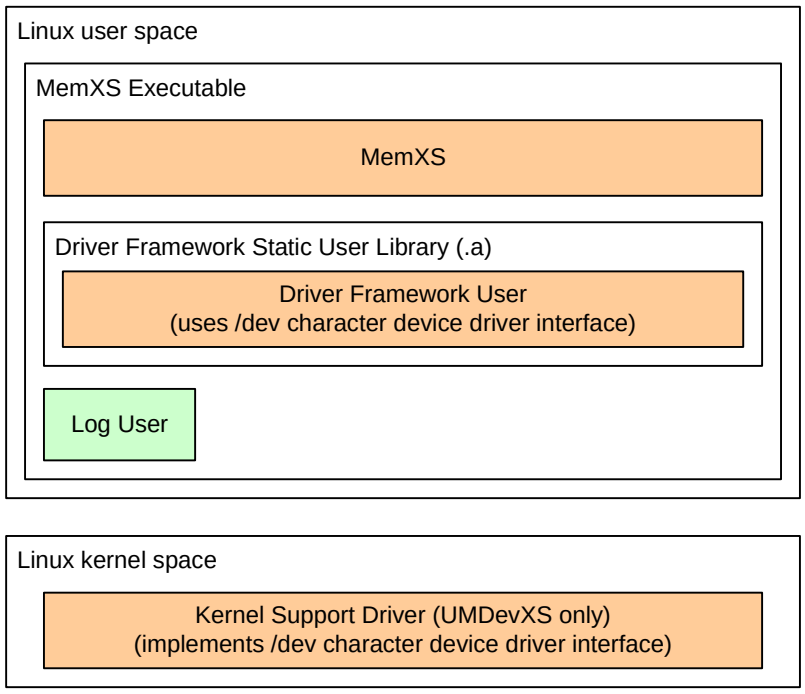


Legend

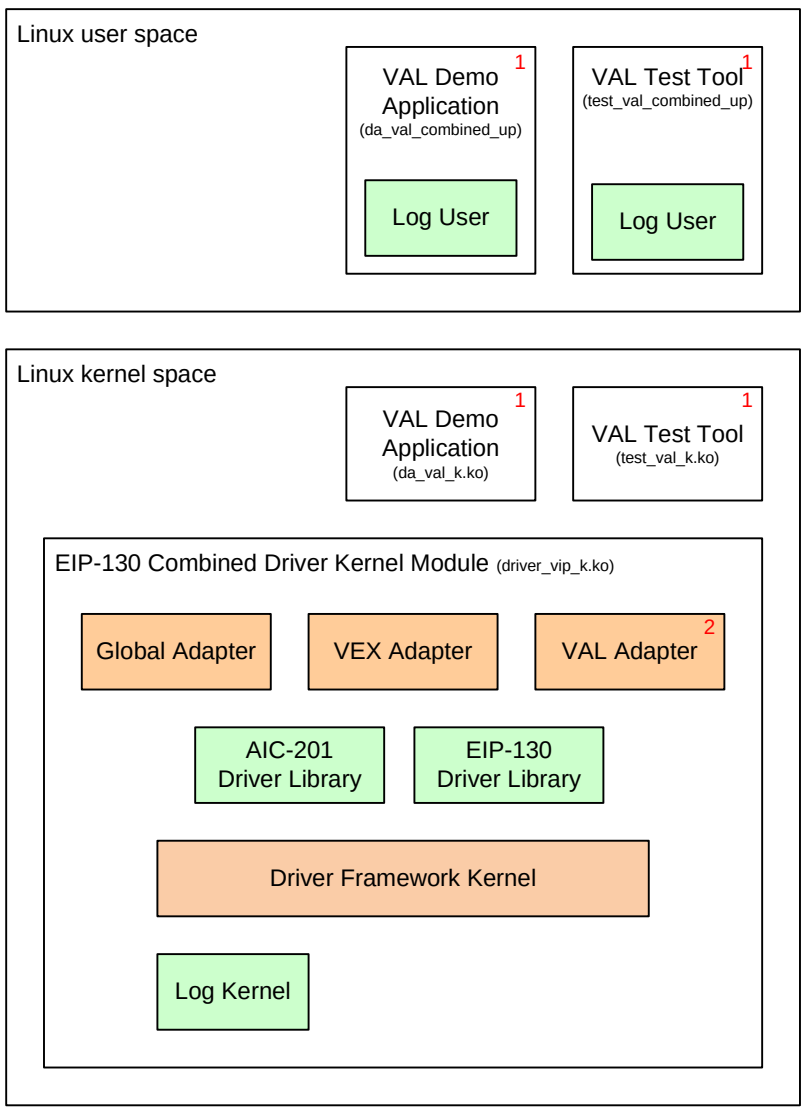
- API — Driver External Interface
- API — Driver Internal Interface
- Implementation Module
- Sub-module
- "API" = Interface only
- Dependency
- Implementation

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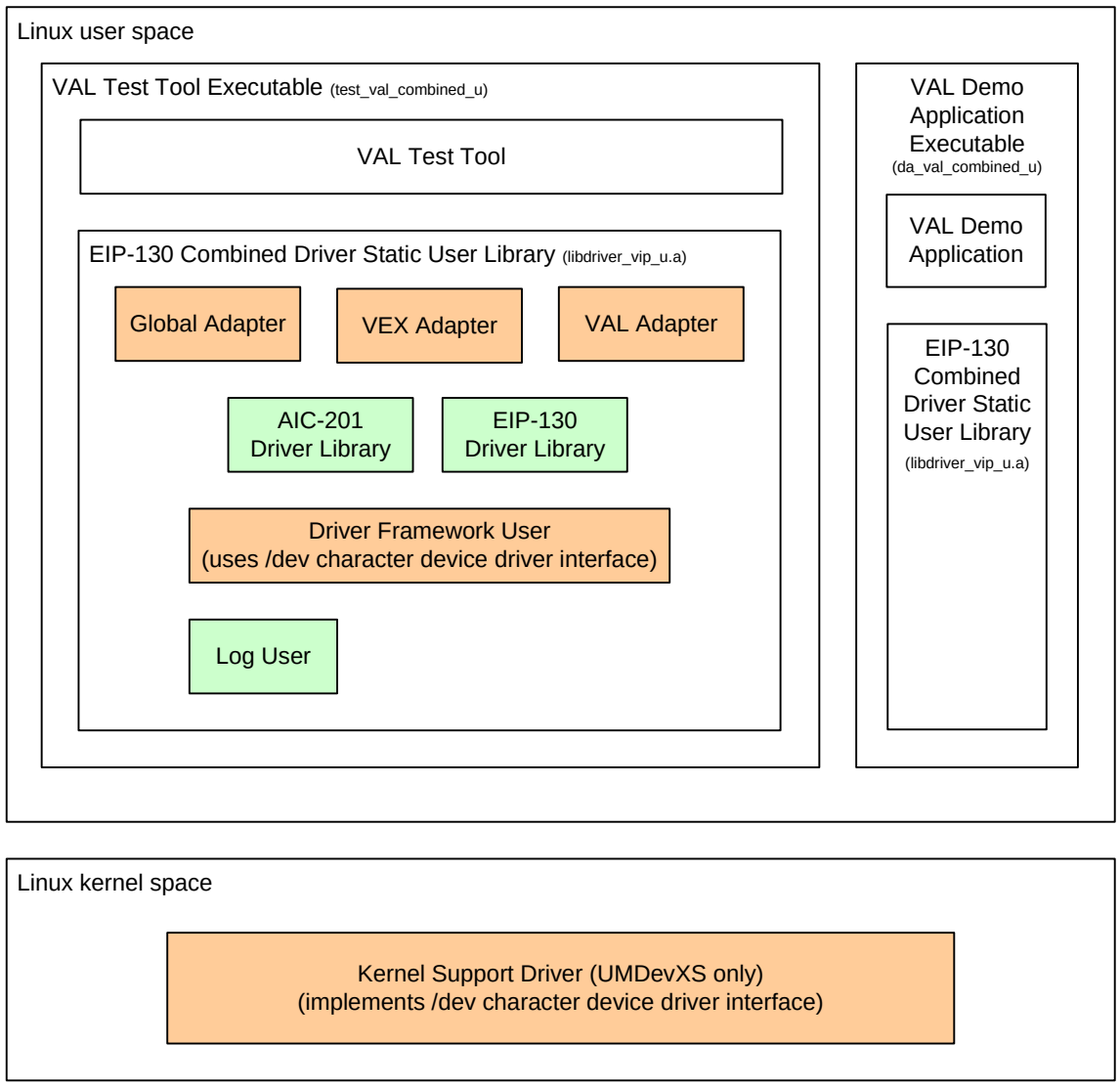
MemXS & Driver Framework User-Space Deployment



EIP-130 Combined Driver Kernel-Space Deployment



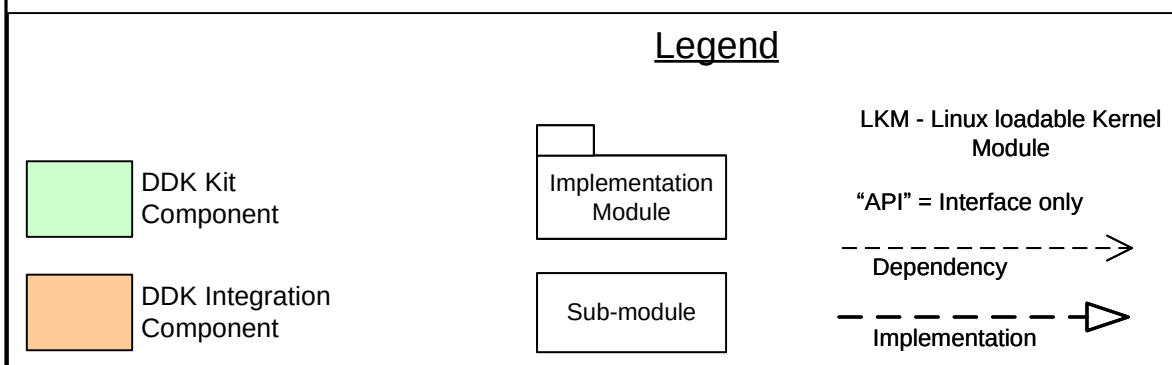
EIP-130 Combined Driver User-Space Deployment

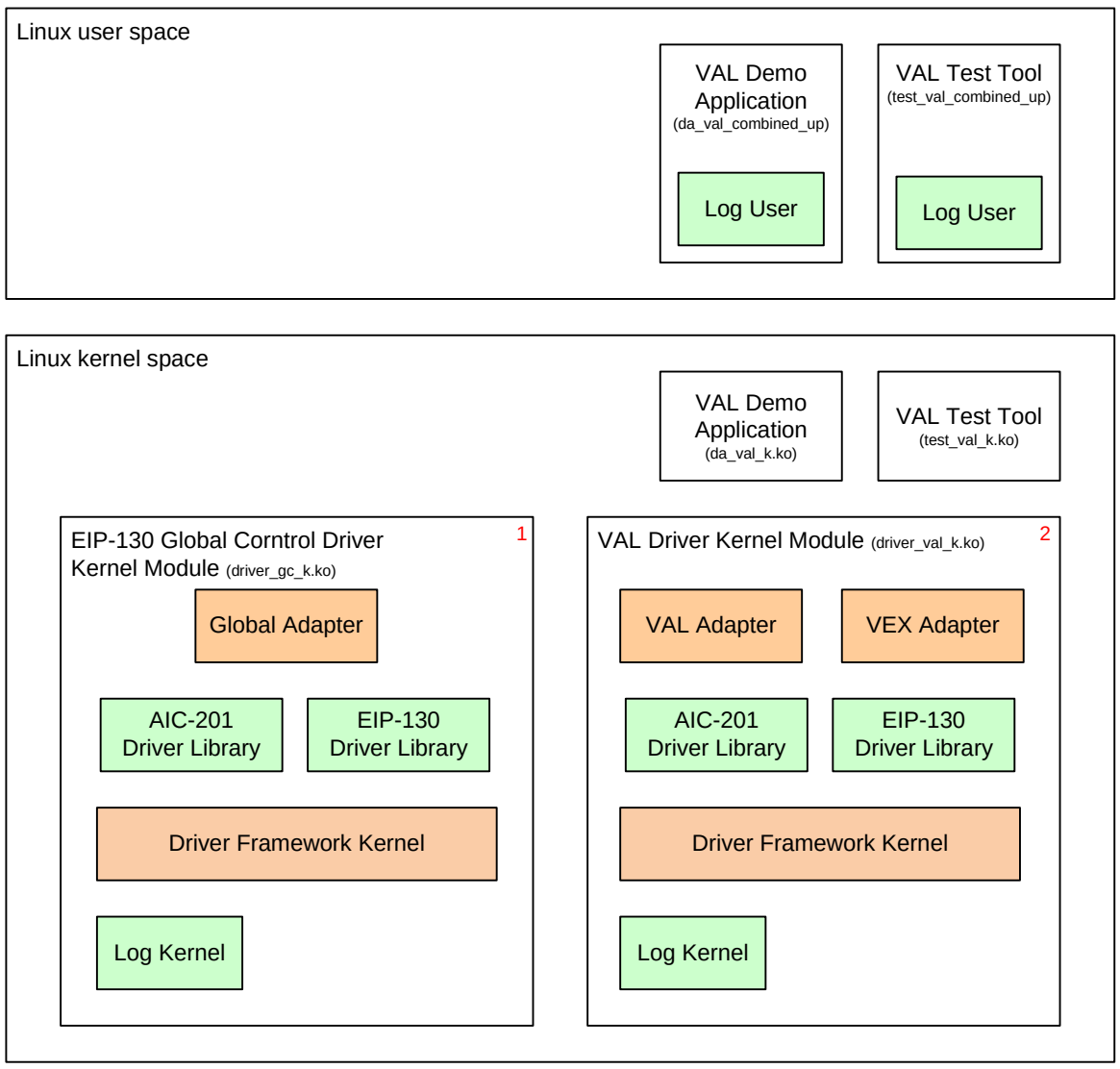


1 Note that only one of these applications (VAL Demo or VAL test tool in user space or kernel space) can be executed at a time.

2 The VAL API is directly accessible for kernel space applications. However, for user space applications it is proxied via VEX Proxy/ Stub, which is not shown.

Legend





1 One instance system wide.

2 One instance per Host.

